Document 00 – ATAI 8WD Master Specifications

Founder: Amirbahador Atai
Version: Gen-1 Engineering Baseline
Language: English

---

1. Executive Technical Overview

The ATAI 8WD platform is a next-generation, ultra-heavy airborne mobility architecture integrating buoyant lift, hybrid propulsion, multi-layered survivability, and modular structural separation. Designed as a disruptive aerial megaplatform, ATAI 8WD combines perimeter helium buoyancy ($72{,}000\,\mathrm{m}^3$), nine concealed high-thrust vertical propulsion units, and a three-layer vertical architecture known as MCVSS (Multi-Core Vertical Structural System). This document defines the top-level engineering baseline for all subsequent subsystem specifications (Documents 01 99).

Mission roles include heavy aerial logistics, amphibious deployment, remote-region access, and emergency mega-evacuation capabilities. The platform supports safe landing on ground or water, long-duration hovering, and survivability under multiple failure scenarios.

---

2. Mission and Operational Concept

Primary missions:

- Ultra-heavy aerial transport (20 60 tons payload envelope)
- Disaster-relief deployment with autonomous access to isolated zones
- Near-silent low-frequency mobility through concealed propulsion
- Amphibious landing and extended water-surface operations
- High-reliability personnel evacuation under degraded conditions

Operational envelope:

- Nominal altitude: 0 2,500 m AGL
- Cruise speed (air-assisted): 60 110 km/h
- Hover endurance: dependent on hybrid turbine output and energy storage
- Water-landing endurance: unlimited with active ballast management

Environmental tolerance:

- Side wind resistance up to 40 knots due to hybrid buoyancy + thrust vectoring
- Operation under light icing conditions with thermal anti-icing surfaces

---

3. Vehicle Architecture and Layer Logic

ATAI 8WD is constructed from three vertical layers:

Layer 1 – Safe Cockpit Pyramid (30 × 30 m)

- Reinforced composite-titanium monocoque
- Independent avionics stack and power module
- 14-day autonomous survival systems
- Capability for independent parachute-based descent

Layer 2 – Mid/Platform Structural Deck

- Power distribution junction
- Access/escalation shafts
- Hybrid turbine-generator mounts
- Distributed load-path grid structure

Layer 3 – Upper Buoyancy & Propulsion Layer

- Perimeter-Deep helium tanks (20 m height edges, 10 m center)

- Concealed 9-fan propulsion matrix
- Thermal-shielded envelope for helium stability
- Structural crown for aerodynamic load absorption

---

4. Structural Configuration – MCVSS

The Multi-Core Vertical Structural System (MCVSS) integrates: - One armored central core shaft (primary load path) - Four auxiliary perimeter access shafts (secondary load paths) - Cross-layer monocoque bonding and composite rib reinforcements - Torsional stiffness increased by factor 2.5 versus single-core configurations

Structural materials:

- Aluminum-Lithium 2195 for longerons and frames
- Carbon fiber (T800/HMJ) for outer skins
- Titanium Grade-5 for load-bearing fasteners and safety-critical nodes
- Multi-layer ceramic armor shielding for cockpit layer

---

5. Helium Buoyancy System

Buoyancy architecture uses a Perimeter-Deep configuration: - Total useful volume: $72{,}000\,\mathrm{m}^3$ - Net lift: approx. $60{,}000\,\mathrm{kg}$ after structural compensation - Natural pressure-equalization channels embedded between tank segments - Rapid isolation valves for puncture control - Double-layer inner envelope preventing diffusion under solar heating

Thermal management:

- IR-reflective film reducing solar heating by 40%
- Passive stratification baffles stabilizing buoyancy in tilt scenarios

---

6. Propulsion & Hybrid Energy System

The system consists of nine high-thrust vertical fans, diameter $9.6\,\mathrm{m}$ each, concealed inside upper structure for reduced acoustic signature and enhanced safety.

Hybrid turbine/generator system:

- 2Ù high-efficiency turboshaft generators
- Power output range: 2 5 MW combined
- Energy buffer: lithium-sulfur battery pack integrated in Layer 2
- Redundant HVDC buses ($\pm 850\,\mathrm{V}$) feeding propulsion fans

Cooling and thermal management:

- Liquid cooling loop with 3-stage radiator system
- Emergency passive cooling for battery integrity

---

7. Safe Cockpit and Survival Systems

Cockpit architecture:

- Armored multi-layer shell with blast-resistant composites
- Redundant avionics (Type-A and Type-B independent systems)
- Life-support for 14 days including CO scrubber, O generation, and water recycling
- Navigational independence using inertial reference + encrypted GNSS

Emergency capabilities:

- 16 ballistic parachutes for full-capsule descent
- Autonomous separation of upper layers under catastrophic conditions
- Crush-zone landing legs (12 hydraulic dampers)
- Fire-resistant interstitial layers

8. Access, Egress, and Internal Logistics

Vertical access system:

- Central armored shaft (primary circulation)
- Four auxiliary perimeter stair modules
- Multi-path evacuation allowing 90-second egress target under emergency

External access:

- Four deployable access bridges for ground boarding
- Water-landing access ramps with self-balancing stabilizers

Logistics:

- Internal cargo elevator (15 20 tons)
- Modular attachment points for mission-specific payloads

---

9. Amphibious Capabilities

ATAI 8WD is capable of stable water operations due to: - Monocoque hull-shaped underside - Multi-cell ballast control with water intake and ejection - Floatation redundancy independent of helium envelope - Anti-corrosion composite layers protecting hull integrity

Water landing profile:

- Impact dampening using hydraulic legs
- Automatic leveling before surface contact
- Transition into low-energy operational mode after splashdown

---

10. Performance, Payload, and Requirements Baseline

Estimated specs:

- Gross platform mass (with helium): 35 45 tons
- Net lift: 60 tons
- Operational payload capacity: 20 30 tons (Gen-1 target)
- Energy reserve endurance: 3 6 hours active flight (hybrid mode)
- Water-landing indefinite endurance with ballast balancing

Interface to subsystem documents:

- Structural load paths   Document 29
- Buoyancy analysis   Document 12
- Propulsion system spec   Document 40
- Electrical HVDC architecture   Document 55
- Safety & emergency design   Document 70

---

11. Development Roadmap

Phase 1 – Concept Verification (2026)

- Mass budget freeze
- Buoyancy prototype testing

Phase 2 – Engineering Definition (2027)

- CAD master model
- CFD and structural FEA

Phase 3 – Prototype Build (2028–29)

- Ground testing + water landing tests
- Hover flight tests

Phase 4 – Certification Path (2030+)

- Non-standard aerial platform certification program
- Safety verification for multi-layer separation

---

12. Key Engineering Risks

- Helium diffusion under long-duration solar exposure
- Fan-induced vibration coupling with buoyant envelope
- Hybrid turbine cooling limits during hover
- Extreme wind loads during water landing

Mitigation strategies include active thermal control, rib reinforcement, optimized thrust vector control, and dynamic ballast algorithms.

End of Document 00 – Master Specifications
Founder: Amirbahador Atai